

Michael Bobak

Summary

Research-Programmer with more than 30 years of experience in the IT industry; with extensive knowledge in building skills around Semantic-Web/Linked-Data, and Knowledge-Engineering and ML; and knowledge of Software Development Life Cycle (SDLC).

Michael's technology expertise includes–

- Research-Programmer starting with (bio)physical-science simulation, adding AI study and years of Knowledge Engineering work as well (in: .com/.edu/.gov/.mil)
- Focusing on Knowledge-Based aids, for process improvement to teaching
- Using AI: Ontology, Knowledge-Representation and Reasoning, Rules, Kn-Acq, NLP, ML

Exceptional analytical and problem-solving skills, a good team player with excellent interpersonal and communication skills.

Education: Bachelor of Science in Physics and Master of Science in Biophysics & Comp-Biology with AI from University of Illinois at Urbana-Champaign

Professional Experience

Pivot Path Solutions (Remote)

Duration: 2026 – present

Responsibilities	▪ Worked as SME for a DARPA digital-twin pitch & suggested a general path fwd incl semantics
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xendat.com/vrtx.com, Remote

Duration: 2023 – 2025

Role	Ontology Data Developer/Analyst (Integrations)
Responsibilities	▪ Contracted to a pharma's Ontology Group using: Python, SPARQL, Triple-Stores, (web) Protege, Semantic-Web-Libs, and some SHACL ▪ Wrote new libs used for both a Semantic-ETL and for an automated QA/QC, aiding Ontologist & BizDev, incl several apps

National Center for Supercomputing Applications, Urbana, IL

Duration: 2019 – 2023

Role	Sr. Research Software Engineer
Responsibilities	▪ Carried my background to earthcube.org and other grants ▪ Focused on semantics/metadata search, with some NLP & sim. EarthCube's GeoCODES: Crawl data repositories json-ld for extended schema.org, and provided faceted search, related datasets, and transition to notebooks ▪ Built up related semantic-web python libs, use of triple-stores. FAIR data principles

	▪ Use of triplestores, SPARQL, and some SHACL
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AlohaHealthNet, Remote

Duration: 2017 – 2025

Role	Sr. Knowledge-Engineering
Responsibilities	▪ Advised early-stage startup built on the topic of my UCSF.edu research ▪ Matched EHR to get people into clinical trails

Agribile/Nutrien Champaign IL

Duration: 2017 – 2018

Role	Sr. Software Engineer
Responsibilities	▪ Planned & guided reworking the main simulation, documentation, & ML/verification which helped with its acquisition

Freelance Consultant, San Francisco, CA

Duration: Jul 2011 – 2017

Responsibilities	▪ Worked as an ontologist for osthus.com on aligning bio/pharma ontologies to BFO to annotate masses of data in HDF5 files, for the allotrope.org ▪ Worked with IDEO on their systems integration issues that could be aided by Knowledge-Graph for info refine/cleanup ▪ Worked with the Siemens Web of Things research group on use of SemWeb+IoT for adaptable manufacturing ▪ Advised with a variety of start-ups in understanding AI tech, including Fashion start-up that would track unstructured blog info to surface trends - Sports startup thewhytehousegroup.com needed dbpedia search ability ▪ Developed ideas to take my UCSF research and fuse it with the Patient Data Mining Cluster that was developed by the UCSF Head of Research Computing and a PhD student, which has now been submitted for a patent ▪ Worked with UCSF in Psychology Department understand how to apply NLP and graph relation insights into an app they developed called Prime for schizophrenic young adults, with application to depression management as well ▪ Helped with an assisted eco-sim/modeling environment in Lisp ▪ Continued to build skills around ML, Semantic-Web/Linked-Data, and Knowledge-Engineering ▪ Done Coursera courses: Data Analysis, Data Science (with distinction), Machine Learning (with distinction), Discrete Optimization (audit) - open-HPI courses: Semantic Web, Knowledge Engineering, Knowledge-Engineering with Semantic Web technology, Linked-Data-Engineering - Stanford courses: Design Thinking
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Apollo Education Group, San Francisco, CA

Duration: Oct 2010 – Jul 2011

Role	Architect, Adaptive Learning Platform
Responsibilities	▪ Conceptually annotated study material and tests for automated remediation, instrument classroom to learn from using: Hadoop, Lisp, KM

UCSF San Francisco, CA

Duration: Sep 2007 – Oct 2010

Role	Programmer/Analyst III
Responsibilities	▪ Performed Medical-Informatics research (relating to clinical-trials) in Lisp/KM, and Natural Language Processing in Java; paper with Stanford group; ontology dev/using: Lisp, KM, Match conceptually annotated patient EMR against a concept based in/ex-clusion criteria annotated from ClinicalTrials.gov to see which trials they might be eligible for, via semantic-search

Freelance Knowledge Engineer/ Research-Programmer

Duration: Feb 2001 – Sep 2007

Summary	mindbox.com 3/02-10/02. (used Art*Enterprise) See: Ocwen_Mindbox
Responsibilities	▪ Worked up to half-time for cas.dis.anl.gov 5/03-5/04 (Java Simulation) ▪ Worked full-time 8/03--05(verizon) labs.gte.com, Model-Based-Diagnosis on a national scale. (Art *Enterprise) See: aaai.org/Papers/IAAI/1996/IAAI96-287.pdf ▪ Involved in Bioinformatics/control contract 11/04-12/05 (CLIPS&Protege.stanford.edu/Java/DB) Control of perfusion pumps on light microscope sample, monitoring incl. Machine-vision, Bio-ontology/reasoning/Kn-mngt for the experiment setup. & Grant proposal work ▪ Worked for CME.com 2/06-06/06 (re)organizing trade-data validation code. (CLIPS/Jess) ▪ Involved in Signal-Processing/Machine-Learning (startup) 06/06-(Lisp/etc.) ▪ Involved in Hospital Informatics/Machine-Learning ghx.com 02/07-05/07-(Lisp), Machine Learning speedup for financial-scientific (Lisp)

UIUC Knowledge Based Systems Lab, Urbana, IL

Duration: Jun 1998 – Feb 2001

Role	Senior Research Programmer
Responsibilities	▪ Organized many levels of a very large knowledge-based simulation projects. Brought over 18 programmers together to deliver a coherent product. Ran weekly group meetings, down to help solving any problem ▪ Involved in hiring, demo, design, install trips, prototyping to lead project direction.

	Taught group of 6 how to use a Rule-Based-shell for a reasoner-rewrite in Art*Enterprise Projects included: Simulation-based, Intelligent Tutoring System (ITS) & Real- Time control system. Being used in classroom, real life testing, presented at IAAI99 'Automated Instructor Assistant for Ship Damage Control' - The system teaches Navy officers how to save a simulated ship in crisis. A variant was developed to catch real-time crisis conditions and suggest solutions www.dwilkins.org/members.htm
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Brightware, Chicago, IL

Duration: Oct 1996 – Jun 1998

Role	Knowledge Engineer
Responsibilities	- Helped develop and install their very first product (intelligent email reply) - Worked between developments and consulting. Helped on several Knowledge-Based business applications. Helped with several deployed Knowledge-Based business applications (i.e. financial: mortgage, web based job finder). w/Art*Enterprise See: www.brightware.com/eservice_solutions & later 1/2 year for the new version of the company: Mindbox

Institute of Learning Sciences, Evanston, IL

Duration: Feb 1996 – Aug 1996

Role	Lead Programmer/Analyst
Responsibilities	- Wrote Lisp code (mainly GUI) for Qualitative Research Group. Learned more about Qualitative/Quantitative Simulation, Model-Based Reasoning, Intelligent-Tutoring-Systems, and general Lisp programming - See: www.qrg.northwestern.edu/projects/NSF/Cyclepad/aboutcp.htm

Argonne National Lab (EAD and DIS groups), Argonne, IL

Duration: Feb 1993 – Feb 1996

Role	Software Engineer
Responsibilities	- Wrote fielded Expert System by myself at the end of grad-school in Lisp rule-shell then CLIPS Prototyped communication and control of distributed simulation. W/CLIPS PVM etc. - Done agent wrapping of simulations with CLIPS+PVM, to describe then mix and match them. Also used C++, Smalltalk, FORTRAN with PVM; Other work as needed. Algo/Viz/Etc. Written up in a book about innovative distributed object application. See: www.dis.anl.gov/DEEM/HLAsim www.dis.anl.gov/DEEM/DIAS/diaswp.pdf Later some work for new subgroup of dis: cas.dis.anl.gov

Educational Background

- University of Illinois, Urbana-Champaign, US
 - o MS – Biophysics & Comp-Biology with AI, Research Assistant /Research Programmer, Year 1993
 - o BS – Physics, BS – Biophysics - Dept. distinction Research Programmer(1/2 time w/local lab), Year 1988

Professional Organizations

- AAAI (Association for the Advancement of Artificial Intelligence) life-member, since uni
- IEEE (Institute of Electrical and Electronics Engineers) & Computer Society 10 years

Others

- Other Groups: meetup.com, linkedin-groups
- ID: orcid.org/0000-0003-2357-5918 wikidata.org/wiki/Q104512704
- Papers: scholar.google.com/&q=michael+bobak